Exercise 1: (**6 Points**) Decide for each of the following statements whether they are **TRUE** or **FALSE**.

a) In a multiple linear regression model, adding a variable which is not correlated with the dependent variable will not affect the unbiasedness of the OLS estimator, but it may affect its variance.
☐ TRUE ☐ FALSE

b) In a linear regression model, minimizing the sum of the absolute deviations results in the same coefficient estimates as minimizing the sum of the squared deviations.
☐ TRUE ☐ FALSE

c) In a linear regression of people's income onto their years of professional experience, adding dummies for the weekday a person was born on as regressors can be expected to lower the R^2.
☐ TRUE ☐ FALSE

d) When the design matrix $\boldsymbol{X}$ does not have full column rank, the OLS estimates still exist and are unique as long as the error variance is constant.
☐ TRUE ☐ FALSE

e) A best linear unbiased estimator (BLUE) is "best" in the sense that there is no other linear unbiased estimator with a lower variance.
☐ TRUE ☐ FALSE

f) In a simple linear regression, the slope estimator $\hat{\beta}_1$ is always uncorrelated with the intercept estimator $\hat{\beta}_0$.
☐ TRUE ☐ FALSE

g) When testing $H_0 : \beta_j = 0$ versus $H_1 : \beta_j \neq 0$ in a linear regression model, the t-statistic for β_j follows a t-distribution with $n-k-1$ degrees of freedom under H_0, where $k+1$ is the number of estimated parameters.
☐ TRUE ☐ FALSE

h) Consider a logit model where you regress y_i onto $1, x_{1,i}, x_{2,i}, \ldots, x_{k,i}$ to obtain $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_k,$. An increase of 1 in $x_{j,i}$ is interpreted as an increase of $\hat{\beta}_j$ in $P(y_i = 1)$.
☐ TRUE ☐ FALSE

i) The F-statistic for testing $H_0 : \beta_1 = -\beta_2 + \beta_3$ in a linear model with $k \geq 3$ predictors plus an intercept has an F-distribution with $(3, n-k-1)$ degrees of freedom under H_0.
☐ TRUE ☐ FALSE

j) Multicollinearity in the design matrix $\boldsymbol{X}$ can inflate the variance of the OLS coefficient estimators.
☐ TRUE ☐ FALSE

k) If a confidence interval for a slope coefficient β_j does not contain zero, we fail to reject $H_0 : \beta_j = 0$.
☐ TRUE ☐ FALSE

l) The OLS estimator in a linear regression is equivalent to the maximum likelihood estimator under the assumption of independent and identically normally distributed errors.
☐ TRUE ☐ FALSE

Exercise 2: (**8 Points**) You want to examine the effect of a person's age, education (no degree, high school degree or college/university degree) and place of birth (inside the US or outside the US) on their hourly wage. To this end, you have sampled randomly from all employed people in the US (both full-time and part-time) and want to use a linear regression model which you estimate by OLS.

a) (3 Points) Provide an adequate linear model equation that incorporates all of the above explanatory variables, including an intercept. Formulate it such that you can estimate this exact equation by OLS.

b) (2 Points) Explain briefly why adding the following regressor to the equation might be a sensible idea:

$$\widetilde{age}_i^2 := (age_i - 48)^2,$$

where age_i is the age of person i.

c) (1 Point) Would you expect the OLS estimate of the coefficient for $\widetilde{age}_i^2$ to be positive or negative? Explain your choice.

d) (2 Points) A colleague suggests adding many more functions of the age_i variable, e.g. $(age_i - 48)^3$, age_i^4 or $\log(age_i)$ because it will improve the fit of your model. How would you decide whether you should do so or not? State two methods.

Exercise 3: (**8 Points**) Given the following R code and its corresponding output, please answer the questions using the information provided here.

```
> lm <- lm(medv ~ crim + nox + dis + rad)
> summary(lm)

Call:
lm(formula = medv ~ crim + nox + dis + rad)

Residuals:
    Min      1Q  Median      3Q     Max
-16.802  -5.152  -2.127   2.604  31.084

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  48.38458    [[A]]    13.591  < 2e-16 ***
crim         -0.25959    0.05302  -4.896 1.32e-06 ***
nox         -36.99122    5.25574  [[B]]  6.44e-12 ***
dis             [[C]]    0.26423  -3.796 0.000165 ***
rad          -0.06165    0.05983  -1.030 0.303290
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: [[D]] on 501 degrees of freedom
Multiple R-squared:  0.2583,Adjusted R-squared:  0.2524
F-statistic: 43.62 on 4 and 501 DF,  p-value: < 2.2e-16

> sum(lm$residuals^2)
[1] 31682.02
```

a) (2.5 Points) Reproduce the values of `[[A]]`,`[[B]]`,`[[C]]` and `[[D]]` in the above regression table. Write down the fitted regression formula of the model.

b) (2 Points) Calculate a 99% confidence interval for the coefficient of the variable `nox` β_{nox}. The confidence interval is supposed to be centered around the estimated coefficient $\hat{\beta}_{nox}$. At the 1% significance level, would you reject the null hypothesis $H_0 : \beta_{nox} = -30$ given the alternative $H_1 : \beta_{nox} \neq -30$?
Note: You can find the correct quantile on the last page.

c) (2 Points) You want to test the joint null hypothesis $H_0 : \beta_{crim} = 3\beta_{rad} - 0.1,\ \beta_{nox} = -40$. Express the null hypothesis as $C\boldsymbol{\beta} = d$, i.e. identify C and d with $\boldsymbol{\beta} = (\beta_0, \beta_{crim}, \beta_{nox}, \beta_{dis}, \beta_{rad})'$. What is the number of linearly independent restrictions r?

d) (1.5 Points) Let $\hat{\boldsymbol{\varepsilon}}'_{H_0}\hat{\boldsymbol{\varepsilon}}_{H_0} = 32333.15$. Calculate the F-statistic and decide whether you would reject the null hypothesis from the previous task at a significance level of 5%.
Note: You can find the correct quantile on the last page.

Exercise 4: (**8 Points**) Answer the following questions.

a) (1 Point) Explain why a linear regression model is not appropriate to model a binary dependent variable and why you should instead use a logit, probit or similar model.

b) (2 Points) Suppose you have a linear regression model

$$y_i = \beta_0 + \beta_1 x_{1,i} + \ldots + \beta_k x_{k,i} + \varepsilon_i.$$

Explain the method of ordinary least squares (OLS) estimation and show the steps necessary to obtain the estimators $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$. It is not necessary to explicitly calculate them; it suffices to show the mathematical approach.

c) (2 Points) Explain what the intuition behind information criteria is. Explain why their design makes them a good tool for model selection. You can use any information criterion you want as an example.

d) (2 Points) Consider the linear regression model

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \varepsilon_i.$$

You want to test the null hypothesis $H_0 : \beta_1 = \beta_2 + 1$ via an F-test. One way to do so is to first estimate the model without restrictions and then again under the null hypothesis to obtain SSE and SSE_{H_0}. Incorporate the null hypothesis into the model equation to obtain a restricted model that you can estimate by OLS directly.

e) (1 Point) You are analyzing the relationship between the yearly revenue of a company and the number of employees it has. When plotting your data set, you notice that the variation in revenue grows as the number of employees grows. You want to use a linear regression to estimate the effect of the number of employees on revenue. Which impact does this phenomenon have on the OLS estimate in terms of bias and efficiency?

Quantile tables

Table 1: Quantiles for some t-distributions

df/quantile	0.9	0.95	0.975	0.99	0.995
3	1.6377	2.3534	3.1824	4.5407	5.8409
7	1.4149	1.8946	2.3646	2.9980	3.4995
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123
25	1.3163	1.7081	2.0595	2.4851	2.7874
41	1.3020	1.6819	2.0180	2.4184	2.6980
49	1.2990	1.6766	2.0095	2.4048	2.6799
50	1.2987	1.6759	2.0086	2.4033	2.6778
100	1.2901	1.6602	1.9840	2.3642	2.6259
250	1.2849	1.6510	1.9695	2.3414	2.5956
501	1.2832	1.6479	1.9647	2.3338	2.5857

Table 2: Quantiles for some F-distributions with fixed $df = 501$

r/quantile	0.9	0.95	0.975	0.99
$r = 2$	2.3132	3.0137	3.7162	4.6478
3	2.0948	2.6227	3.1422	3.8209
4	1.9561	2.3897	2.8114	3.3568
5	1.8588	2.232	2.5918	3.0539